

When to Intervene: Operationalizing AARC Deficit Labels as a Reward Signal for AI-Assisted Pre-Mediation

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Abstract

This paper presents a data-focused contribution to preference-based reinforcement learning for AI-assisted pre-mediation. We operationalize the AARC framework as turn-level annotation targets for conflict dialogue, construct a 100-dialogue synthetic corpus using a two-agent persona-scenario pipeline, and develop an annotation schema achieving mean $\kappa = 0.671$ against a human gold standard. The reward model training phase was not completed within the project timeline; results cover annotation development in full, including schema iteration (v1: $\kappa = 0.401 \rightarrow$ v2: $\kappa = 0.671$), annotator comparison across three frontier models, and a diagnostic failure analysis for two of four AARC components. The annotation contribution stands independently: a reliable schema with diagnosed failure modes is a more informative result than a trained model with unvalidated inputs.

0.1 Introduction

This paper addresses a structural limitation in current AI-assisted dialogue management: the tendency to respond to the propositional surface of a dispute while its relational, procedural, and identity-based dimensions remain unaddressed. The task is turn-level deficit detection using the AARC framework (Hoben, 2020) as a reward signal for preference-based reinforcement learning in AI-assisted pre-mediation — one-on-one sessions in which an AI assistant works with a single conflict party before any joint engagement with their counterpart.

The research track is *Research*; the focus is *Data*. The choice of a data focus reflects where the project’s substantive contribution lies: the primary work accomplished is the construction and annotation of a synthetic conflict corpus, together with the schema development required to make that corpus reliable. The reward model training that the corpus enables was not completed within

the project timeline; the data track is appropriate because the annotation contribution is the primary deliverable.

The motivating problem is what conflict resolution theory calls the *content trap* — the tendency to address only the propositional surface of a dispute while relational, procedural, emotional, and identity-based dimensions remain unresolved (Wilmot and Hocker, 1997). FPO (Frictive Policy Optimization), the computational dialogue management system this project extends, is a well-specified instance of this failure: its friction functional decomposes epistemic risk into five measurable components but contains no moves for acknowledgment, face-repair, or process affirmation (Pustejovsky and Krishnaswamy, 2025). A system that cannot detect when a user is not yet ready to receive calibration will deliver content-level correction into conditions that make it counterproductive.

The AARC model (Hoben, 2020) — Acknowledgement, Agency, Reciprocity, Clarity — provides the theoretical framework for detecting those conditions at the turn level. The central claim of this project is that AARC deficit labels can be operationalized as observable, turn-level annotation targets with demonstrable inter-annotator reliability, and that such labels would produce a richer reward signal than FPO friction scores alone. This paper establishes the first part of that claim; the second remains an empirical question for the training phase.

The sections that follow situate the work in the relevant literature (§2), describe the corpus construction methodology and annotation pipeline development in full (§3), specify the planned training pipeline (§4), report annotation results and pipeline state at submission (§5), analyze what worked and what did not (§6), and conclude with a reframing of the contribution scope (§7).

0.2 Related Work

FPO as computational baseline. Frictive Policy Optimization reframes alignment as a problem of epistemic control: rather than optimizing for helpfulness or stated preferences, it asks when an agent should intervene to regulate the quality of belief evolving over a dialogue trajectory (Pustejovsky and Krishnaswamy, 2025). Its friction functional decomposes epistemic risk into uncertainty miscalibration (*Unc*), logical contradiction (*Contr*), safety hazard (*Haz*), value conflict (*ValConf*), and expected information gain (*InfoGain*). This architecture is well-suited to epistemic correction but has no vocabulary for the relational, identity, and procedural dimensions that conflict resolution theory identifies as primary drivers of impasse.

AARC framework. The AARC model (Hoben, 2020), grounded in practitioner conflict resolution work on dignity (Hicks, 2011; Suskind and Field, 1996), identifies four components a practitioner must address: Acknowledgement, Agency, Reciprocity, and Clarity. It is a practitioner heuristic without formal empirical validation. Shaikh et al. (2024) provide the closest computational precedent: they demonstrate that the Interest-Rights-Power framework can be used to condition LLM outputs in a conflict-training context, producing measurable improvements in subsequent conflict behavior. That work is training-oriented and single-framework; operationalizing a four-component deficit schema as turn-level annotation targets is this project’s extension.

Preference-based RL. The training framework follows the standard preference learning formulation: the Bradley-Terry model (1952) serves as the pairwise comparison model, as operationalized in RLHF by Christiano et al. (2017) and scaled by Ouyang et al. (2022). Policy training via group relative policy optimization (DeepSeek-AI, 2025) is the intended downstream application.

Synthetic corpus generation. DiaSynth (Suresh et al., 2025) provides the closest methodological precedent: an LLM-based synthetic dialogue corpus conditioned on structured personas and scenarios that captures 90.48% of the performance distribution of in-domain naturalistic data. The persona-scenario architecture adopted here follows DiaSynth’s design but extends it with conflict-psychology dimensions necessary to produce reliable annotation targets.

Director architecture. The two-level genera-

tion design — a character-playing generator monitored by a director that checks profile fidelity — has computational precedent in Riedl and Young’s (2010) IPOCL narrative planner, which demonstrates that explicitly balancing character intentionality against author-level coherence produces more coherent character behavior than single-level generation.

Conflict psychology. The persona design draws on Bowlby’s (1973) internal working models (beliefs about self and others), Thomas’s (1976) TKI conflict-handling modes (assertiveness $\times$ cooperativeness taxonomy), Pruitt and Rubin’s (1986) escalation trajectories, and Ting-Toomey’s (1988) face-negotiation theory. These grounding frameworks constrain character behavior in ways that surface-level persona attributes cannot.

0.3 Data

0.3.1 Corpus Construction

The corpus construction methodology makes two design decisions with direct consequences for annotation validity. First, dialogues are conditioned on persona-scenario pairs rather than on AARC deficit categories directly. Prompting for deficit categories — “write a dialogue showing an Acknowledgement deficit” — risks producing stylized instantiations that are too uniform to serve as reliable annotation targets and introduces the AARC vocabulary into the generation context, creating a risk of annotation circularity at the labeling stage. Persona-scenario conditioning allows deficit patterns to emerge from character psychology interacting with situational stakes, rather than being specified as generation targets.

Second, dialogues are generated using a two-agent architecture in which each LLM instance is provided only with its own character profile and the shared scenario brief, not with the opposing character’s profile. A single LLM generating both sides of a dialogue has access to both parties’ goals simultaneously, which tends to produce cooperative exchanges in which misunderstandings resolve readily. Restricting each instance to its own persona replicates the information asymmetry characteristic of real conflict, allowing goal incompatibility and deficit accumulation to arise from character dynamics rather than scripted construction.

Generation uses chain-of-thought prompting following Suresh et al. (2025): each generating in-

stance first reasons through the character’s current activation state before producing the turn text, ensuring that deficit-relevant behavior accumulates as a pattern across turns. LLMs trained with RLHF exhibit a strong prior toward cooperative, de-escalating behavior; generation is framed as a scripted dramatic exercise — each instance as an actor performing a character in a scene — to counteract this prior. A director LLM monitors each generated turn against the character’s declared profile and flags drift toward cooperative defaults before it propagates across the window.

Persona library. The library consists of 20 character profiles, each grounded in nine dimensions: *core needs* (what the character requires to remain engaged); *beliefs about self* and *beliefs about others* (Bowlby’s (1973) internal working models, governing how the persona reads ambiguous moves); *face salience* (following Ting-Toomey’s (1988) face-negotiation theory, the degree to which concession or correction registers as identity threat); *TKI conflict-handling mode* (Thomas, 1976); *behavioral pattern under activation*; *activation profile and storm shift* (Pruitt and Rubin, 1986); and *repair orientation* (what form de-escalation must take to register as genuine). Profiles were constructed through structured prompting against conflict resolution and social cognition literature without naming AARC components in the generation prompts, preventing persona vocabulary from anticipating the annotation targets.

Each persona also carries a *communicative register* specification and a *pragmatic stylistics profile* assigned from a library of six profiles across four surface dimensions: silence and pause handling, filler frequency, turn-taking conventions, and hedging and qualification patterns. Profiles range from low-context direct through high-context reflective, cooperative overlap, formal institutional, casual high-filler, and hierarchical deferential. Cross-assigning stylistics profiles independently of TKI modes produces pairings in which surface register varies orthogonally to conflict dynamics, reducing the cross-dialogue surface uniformity observed in early generation trials.

Scenario library. The library consists of 20 conflict briefs, each defined along five fields: conflict type, relationship context, surface issue, and separate briefs for party A and party B specifying each party’s concrete risks, constraints, and underlying drivers. Scenarios were derived via RAG-assisted review of practitioner conflict resolution manuals,

sampling across conflict domains including workplace, family, commercial, neighbourhood, and institutional contexts.

Dialogue generation. Each dialogue pairs two personas with a scenario brief. Dialogue length is governed by a moderator check applied after each turn: either a rule-based condition (turn limit reached or explicit goal achievement detected) or a dedicated LLM call judging whether natural closure has been reached. Dialogues run to a minimum of eight turns to ensure sufficient coverage under the seven-turn sliding window used at annotation time, and a maximum of twenty turns.

The resulting corpus consists of **100 dialogues, 1,440 turns**, and an average of 14.4 turns per dialogue (min: 8, max: 20), drawn from pairings across the 20-persona and 20-scenario libraries.

0.3.2 Annotation Pipeline — Development and Failures

Schema v1 and its failure. The annotation schema labels each dialogue turn independently for each of the four AARC components — deficit present or absent — using a trailing seven-turn context window. The first IAA assessment, conducted under the v1 schema across 90 matched turns, yielded mean $\kappa = 0.401$ (A: 0.560, G: 0.352, R: 0.379, C: 0.313). Clarity was the weakest component at fair agreement. The principal failure mode was identified as a confusion between deficit signals and repair attempts. The schema’s speaker-attribution principle requires that a Clarity label marks the focal speaker as *proceeding on unverified epistemic ground* — not as articulating an uncertainty gap. An utterance naming an epistemic gap and orienting toward its resolution is a repair move, not a deficit. The deficit signal is structurally opposite: a speaker treating an unverified interpretation as a settled premise. This inversion — between naming a gap and silently crossing it — proved difficult for both human and LLM annotators to track consistently. The error was symmetric: human and model showed the same characteristic pattern, labeling repair-oriented turns as deficit-positive.

Schema redesign. The v1 schema was refactored to introduce explicit repair resolution indicators (distinguishing post-repair confirmation, speaker-as-repairer, and accommodation without genuine re-engagement), backward-looking discriminators for adjacent components cross-documented from both directions, and worked gold standard examples covering deficit trajectories, res-

olution transitions, and boundary cases. The re-design addressed the repair/deficit inversion at the cost of a substantially expanded annotation instrument.

Schema v2 results. Re-annotation of the gold set under the refined schema produced consistent improvement across all four components. Agreement via Claude Sonnet-4-6 stabilized across the final two annotation iterations (139 and 140 turns respectively) at mean $\kappa = 0.671$ (A: 0.767, G: 0.597, R: 0.789, C: 0.530) — a net gain of +0.270 over the v1 baseline, with the largest improvement on Clarity (+0.217).

Annotator evaluation. With a stable gold standard established, silver annotation scaling was attempted using a DSPy-optimized annotator (GPT-4o, BootstrapFewShot) compiled against 15 hand-curated few-shot demonstrations. Evaluation against 100 gold turns yielded mean $\kappa = 0.391$ — approximately the v1-schema baseline and well below the refined-schema ceiling. The Clarity component showed the lowest agreement and a bidirectional error pattern consistent with residual repair/deficit confusion, confirming the capability gap as structural rather than prompt-dependent.

GPT-5.4 was subsequently evaluated against the 140-turn gold set as a diversity control, yielding mean $\kappa = 0.413$. Cross-model agreement between Claude Sonnet-4-6 and GPT-5.4 is $\kappa = 0.340$, confirming that the two models produce genuinely divergent labelings rather than correlated errors — validating the diversity rationale to have a different LLM perform the evaluation on the gold labels than the one which produced the taxonomy. MiPROv2 optimization (light budget: 10 trials, 6 instruction candidates) improved GPT-5.4 macro-F1 from 0.419 to 0.505 on a held-out val set, with Acknowledgement reaching F1 = 0.769. Medium-budget optimization (18 trials) regressed macro-F1 to 0.493, and critically suppressed Clarity prediction entirely — the best-scoring configuration predicted Clarity-absent for all val examples, yielding F1 = 0.000 on that component. This outcome identifies a structural vulnerability in per-example accuracy metrics: with Clarity positive in only 15.7% of turns, an all-absent strategy scores 84.3% label accuracy and the optimizer exploited this imbalance.

A recall-weighted optimization metric and a three-component annotator (Clarity excluded) were subsequently evaluated. The optimized three-label program reached macro-F1 = 0.572 (A: 0.800, G:

0.250, R: 0.667) against an unoptimized baseline of 0.436 — a gain of +0.137. Agency, however, regressed to F1 = 0.250, with 21 false positives against 4 true positives across 18 trials. The boundary between Agency deficit and general conflict distress is not specified with sufficient behavioral precision in the current taxonomy for a prompted model to locate reliably. The pattern is structurally identical to the Clarity failure: the problem is definitional, not a product of metric design or search budget.

These results establish the limit of the DSPy/GPT-5.4 optimization track. Two of four AARC components — Clarity and Agency — cannot be reliably annotated at corpus scale without taxonomy revision. Claude Sonnet-4-6 (mean $\kappa = 0.671$) remains the only annotator meeting the quality threshold for corpus scaling.

0.3.3 Corpus Statistics

Table 1 summarizes corpus composition. Table 2 shows the gold label distribution. Table 3 presents inter-annotator agreement across schema versions and annotators.

Metric	Value
Dialogues	100
Total turns	1,440
Mean turns / dialogue	14.4
Min / max turns	8 / 20
Persona profiles	20
Scenario briefs	20
Stylistics profiles	6
Gold-annotated turns	140
Silver-annotated turns	0 (in progress)
Unannotated turns	1,300

Table 1: Corpus composition.

Component	Positive	Rate
Acknowledgement (A)	32 / 140	22.9%
Agency (G)	15 / 140	10.7%
Reciprocity (R)	18 / 140	12.9%
Clarity (C)	22 / 140	15.7%

Table 2: Gold label distribution (140 turns).

Corpus validity limitations. The dialogue generation pipeline has not been empirically validated against naturalistic conflict data. Suresh et al. (2025) demonstrate that synthetic dialogues generated under analogous conditions capture 90.48% of the performance distribution of in-domain naturalistic data, but that result is domain-general and does not establish fitness for deficit detection in

Condition	Turns	κ (A)	κ (G)	κ (R)	κ (C)	Mean κ
V1: Human vs. Claude Sonnet	90	0.560	0.352	0.379	0.313	0.401
V2: Human vs. Claude Sonnet	140	0.767	0.597	0.789	0.530	0.671
V2: Human vs. GPT-5.4	140	0.534	0.389	0.533	0.196	0.413
V2: Claude vs. GPT (cross-model)	140	0.597	0.272	0.349	0.140	0.340

Table 3: Inter-annotator agreement across schema versions and annotators.

mediation-relevant conflict. The corpus is also appropriately understood as scoped to a specific communicative register: LLMs reproduce dominant Western, educated, metropolitan communicative norms, and the annotation schema encodes this implicitly. Its utility for cross-cultural mediation contexts has not been assessed.

0.4 Methods

0.4.1 Annotation-to-Training Pipeline

The annotated corpus becomes training signal through the following pipeline. Window-level deficit status is derived algorithmically: a seven-turn sliding window is designated deficit-present on a given component if at least one turn within it carries a positive label for that component. The count of positive turns within the window constitutes the severity signal. Annotating at the turn level rather than the window level directly serves two purposes: it produces a finer-grained reward signal, and it maintains consistency between the annotator’s information set and the downstream model’s inference-time information set.

Windows in which at least one component is labeled deficit-present constitute candidate intervention points. At each such point, N candidate interventions are generated from the component-appropriate intervention types defined in the companion taxonomy. Each intervention is scored across all four AARC components — whether it targets the detected deficit and whether it avoids worsening components not currently in deficit — and an aggregate score is computed as the unweighted sum. The group is ranked by aggregate score. Ranking N interventions rather than selecting between two captures preference intensity across the full taxonomy and preserves multi-component structure in the reward signal that a binary preference would collapse.

Pairwise preference pairs are derived from group rankings by sampling contrastive pairs across the ranked list, applying the Bradley-Terry model (1952) as operationalized in RLHF reward model

training (Christiano et al., 2017; Ouyang et al., 2022). The corpus is designed to yield between 5,000 and 20,000 preference pairs.

0.4.2 Reward Model

The reward model is LLaMA 8B fine-tuned on the pairwise preference corpus. Training hyperparameters are to be finalized at training time; architecture and compute environment will be reported at that stage. A held-out agreement threshold of 70% serves as the quality gate before training proceeds. Agreement below this threshold indicates either annotation noise or an insufficiently specified rubric and triggers schema-level revision rather than proceeding.

0.4.3 Baseline

The friction-only baseline is trained on preference pairs derived from the raw FPO friction functional — a composite of submodel scores for Unc, Contr, Haz, ValConf, and InfoGain — without AARC deficit labels. The preference pair structure is identical to the AARC-enriched condition; only the scoring signal differs. This isolates the contribution of AARC labels to the reward signal.

0.4.4 Evaluation

Evaluation metrics are preference prediction accuracy on the held-out set, per-component F1 against the gold standard for deficit labels, and Krippendorff’s alpha across the four binary components (Artstein and Poesio, 2008). The ablation study compares the AARC-enriched model against the friction-only baseline on these metrics. Policy training via GRPO (DeepSeek-AI, 2025) — in which the policy samples G candidate completions per prompt and the reward model scores each — is the intended downstream application and is out of scope for this project.

0.5 Results

The reward model training and ablation study were not completed within the project timeline. This

section reports annotation results and describes the state of the pipeline at submission.

0.5.1 Annotation Results

Table 3 (§3c) presents inter-annotator agreement across schema versions and annotators in full. The headline results are: V1 schema achieved mean $\kappa = 0.401$ across 90 matched turns, insufficient for corpus-scale annotation. The redesigned V2 schema achieved mean $\kappa = 0.671$ across 140 turns, a net gain of +0.270. The gain was consistent across all four components; Clarity improved most (+0.217) and Agency least (+0.245).

Table 4 shows the full component-level breakdown for the primary annotator (Claude Sonnet-4-6 vs. human gold, 140 turns).

Acknowledgement and Reciprocity annotate at substantial agreement ($\kappa \geq 0.75$); Agency and Clarity annotate at moderate agreement. The component-level pattern is consistent with the definitional findings from the DSPy optimization track: Acknowledgement and Reciprocity have operationally specific deficit signals; Agency and Clarity have underspecified boundary conditions.

0.5.2 Pipeline State at Submission

Silver annotation: not yet complete. The full 1,300-turn pass is pending; gold-annotated coverage stands at 140 turns across 13 transcripts. Preference pair generation, reward model training, and ablation study have not been initiated.

0.6 Analysis and Discussion

0.6.1 What Worked

Schema iteration as productive failure. The v1-to-v2 progression is itself a finding: annotation quality for AARC components is substantially improvable through principled redesign of boundary conditions and worked examples. A net gain of +0.270 in mean κ — from fair to substantial/moderate agreement — was achieved through explicit repair resolution indicators and cross-documented boundary case discriminators. The iterative process was not a failure to get it right the first time; it was the mechanism by which the boundary conditions were identified at all.

Acknowledgement and Reciprocity. Both components annotate reliably at corpus scale ($\kappa = 0.767, 0.789$). Their deficit signals are operationalized with behavioral specificity sufficient for a prompted frontier model to generalize: Acknowledgement’s

self-minimizing language, universalizing appeals, and grievance cataloguing are detectable in surface text; Reciprocity’s positional language and other-directed dismissal signals are similarly observable. DSPy optimization on Acknowledgement reached $F1 = 0.800$ with GPT-5.4, within 0.024 of the Claude ceiling — confirming that the operationalization is portable across model families when well-specified.

Persona-scenario pipeline design. The two-agent architecture combined with drama framing produced dialogues with genuine conflict dynamics. Preliminary inspection of generated transcripts shows escalation, entrenchment, and positional hardening that do not appear in single-agent generation at the same conflict intensity. The director successfully flagged and interrupted cooperative drift in early trial runs.

0.6.2 What Didn’t

Agency and Clarity definitional failure. Both components proved resistant to optimization for the same underlying reason: underspecified boundary conditions. For Clarity, the repair/deficit inversion — distinguishing a speaker who names an epistemic gap from one who silently crosses it — was identified in v1 IAA and addressed in v2 redesign, but DSPy’s medium-budget optimization still collapsed to an all-absent strategy on Clarity, exploiting class imbalance. The fix required removing Clarity from the optimization target entirely, which addressed the optimization problem without resolving the underlying definitional one.

For Agency, the boundary between deficit — perceived foreclosure on one’s ability to influence the situation — and general conflict distress or negativity may not be specified with sufficient behavioral precision. Powerless speech markers appear in both states; the discriminating signals are contextual and accumulative in ways the current schema does not fully specify. The optimization revealed this: the recall-weighted metric was specifically designed to prevent all-absent suppression, and Agency had sufficient positive-class training examples, yet the optimizer could not improve Agency precision beyond 0.160 across the full medium-budget optimization run.

Annotator evaluation cost. The substantive evaluation of three annotators across multiple optimization runs — DSPy/GPT-4o, Claude Sonnet-4-6, GPT-5.4 with light and medium MiPROv2 budget — was an unplanned cost. The initial as-

Component	κ	α	P_o	Human+	LLM+	F1
Acknowledgement	0.767	0.767	91.4%	22.9%	25.7%	0.824
Agency	0.597	0.595	90.7%	10.7%	15.7%	0.649
Reciprocity	0.789	0.788	95.7%	12.9%	10.0%	0.812
Clarity	0.530	0.528	85.7%	15.7%	21.4%	0.615
Mean	0.671	0.670	90.9%			0.725

Table 4: Claude Sonnet-4-6 annotation quality (V2 schema, 140 turns, human gold as reference).

sumption was that a well-specified schema would generalize straightforwardly across frontier models. The capability gap between GPT-4o and Sonnet on this task proved to be model-level rather than prompt-level, and confirming this required an evaluation pass that displaced time from the training pipeline.

Pipeline scope. The annotation track — schema design, IAA assessment, schema revision, gold re-annotation, and annotator evaluation across three models — consumed substantially more effort than projected, displacing the RL training phase. This is not a planning failure in the pejorative sense: the IAA work was load-bearing. Proceeding to train a reward model on unvalidated annotation would have produced results of uncertain interpretability.

0.6.3 Why the Annotation Contribution Stands Independently

The core theoretical contribution — operationalizing AARC as turn-level annotation targets with demonstrated inter-annotator reliability — does not require the reward model to be validated. A reliable annotation schema with mean $\kappa = 0.671$, a diagnosed failure mode for two components (Agency and Clarity), and a confirmed quality ceiling for the silver annotator is a more informative result than a trained model whose annotation inputs were unvalidated.

The methodology is reusable and constitutes the foundation the reward model will be trained on. The persona-scenario pipeline, director architecture, and schema design rationale are documented in sufficient detail for replication. The identified definitional failures of Agency and Clarity are actionable: they specify exactly what the next schema revision must address, rather than leaving a diffuse sense that something is off.

There is also a methodological insight with independent value: optimization pressure is a useful diagnostic tool for annotation schema quality. The DSPy track revealed that Clarity’s and Agency’s boundary conditions were underspecified in ways

that IAA metrics alone did not surface — medium-budget optimization exposed the failure modes in a way that reviewing annotation errors manually would have taken substantially longer to reveal.

0.6.4 Evaluation Circularity

The design has a structural limitation that bounds the interpretability of all results: the reward model (when trained) will be evaluated against annotations derived from the same researcher’s judgments on LLM-generated synthetic dialogues. IAA metrics measure label consistency between annotators, not the validity of AARC as a predictor of mediation effectiveness. A gold standard achieved through Claude Sonnet-4-6 agreement at mean $\kappa = 0.671$ establishes that the schema is consistently applicable; it does not establish that the labels correspond to the psychological states the AARC model names, or that interventions scored highly by the resulting reward model would be judged effective by a practicing mediator. External validation — mediator judgments on system-generated interventions, or behavioral outcome measures from controlled conflict sessions — is required before claims about real-world utility can be substantiated.

0.6.5 Forward Plan

Near-term work requires taxonomy revision for Agency and Clarity: clearer deficit boundary conditions, worked examples covering the deficit-versus-adjacent-phenomenon distinction, and a fresh human gold pass under the revised schema. The training pipeline then proceeds: window derivation → intervention generation → preference pair construction (target: 5,000–20,000 pairs) → reward model training (LLaMA 8B) → ablation (AARC-enriched vs. friction-only). The 70% held-out agreement quality gate applies before training proceeds. Policy training (GRPO) is the downstream application and is deferred to follow-on work.

Two corpus validation tracks are also planned. The first is naturalistic validation: systematic comparison of the synthetic corpus against a reference set of naturalistic conflict transcripts, assessing

whether the distribution of deficit signals, escalation trajectories, and stylistic variation in the synthetic data is consistent with what is observed in real interactions. The second is cross-cultural applicability: the current corpus encodes Western, educated, metropolitan communicative norms by construction, and applying the annotation schema to conflicts operating under significantly different conventions of face, silence, and directness — including First Nations mediations and intercultural disputes — requires both register-stratified corpus expansion and consultation with practitioners from target cultural contexts before the schema can be applied.

0.7 Conclusion

This project asks whether AARC deficit labels produce a richer reward signal than FPO friction scores alone. That question remains open. The annotation infrastructure to answer it is now in place, but the training phase was not completed within the project timeline.

What was demonstrated: AARC deficit labels can be operationalized as turn-level annotation targets with reliable inter-annotator agreement for two of four components (A: $\kappa = 0.767$; R: $\kappa = 0.789$), and with moderate agreement for the remaining two (G: $\kappa = 0.597$; C: $\kappa = 0.530$). The schema is substantially improvable through iteration: the v1-to-v2 redesign produced a net gain of +0.270 in mean κ across four components. The definitional failure of Agency and Clarity is a finding rather than just a gap — it names a specific problem (underspecified boundary conditions between deficit and adjacent phenomena) and points to a specific fix.

The Agency and Clarity result also demonstrates that optimization pressure is a useful diagnostic tool for annotation schema quality. Medium-budget MiPROv2 optimization suppressed Clarity prediction entirely and failed to improve Agency precision across all budget trials — exposing definitional failures that IAA metrics alone did not surface.

Two concrete next steps: taxonomy revision for Agency and Clarity, and execution of the RL training pipeline on the annotated corpus once the silver pass and revision are complete. Whether AARC labels add discriminative value over the friction-only baseline is an empirical question; the annotation work makes it a tractable one.

Limitations

The primary limitation is the evaluation circularity described in §6: IAA metrics establish annotation consistency, not construct validity. External validation against human mediator judgments or behavioral outcomes in controlled sessions is required before claims about real-world mediation utility can be substantiated. The synthetic corpus has not been empirically validated against naturalistic conflict data, and its register scope is limited to Western, educated communicative norms. These limitations are structural to the proof-of-concept scope and are deferred to future work.

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